



[WEBSITE PROJECT PROPOSAL]

[Web Development (Introduction)/ WEDE5020]
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1.1 Organisation Overview

Name: UrbanNest Rentals

History: I established UrbanNest Rentals in 2018 after identifying a significant gap in the market for transparent, tech-enabled urban apartment rentals. As a former real estate agent, I had witnessed firsthand how frustrating the rental process was for young professionals — hidden fees, slow responses, and endless paperwork. I started the business with just 12 studio apartments in downtown Johannesburg, managing everything myself from a small office. Over the past eight years, I have grown the company to manage over 150 properties across three major South African cities, and I have built a team of six dedicated staff members who share my vision for modern, hassle-free renting.

Mission Statement: My mission is to provide urban professionals with hassle-free access to quality rental properties through transparent pricing, digital-first processes, and exceptional customer service.

Vision Statement: My vision is to make UrbanNest the most trusted digital rental platform for urban living in Africa by 2030.

Target Audience: I primarily target young professionals aged 25–35, digital nomads, and small families earning between R25,000–R60,000 monthly. From my experience, these users are tech-savvy, mobile-first, and value convenience and speed in their property search. I have designed my services around their expectations.

1.2 Website Goals and Objectives

Specific Goals:

- I aim to increase qualified lead generation by 40% within six months of launch
- I want to reduce my average time-to-lease from 14 days to 7 days through digital applications
- I need to establish my brand credibility as a modern, trustworthy rental platform
- I plan to automate 60% of tenant inquiries through self-service features, freeing up my team for complex cases

Key Performance Indicators (KPIs):

- Monthly website traffic (my target: 10,000 unique visitors by month 6)
- Conversion rate from property view to application (my target: 8%)
- Average session duration (my target: 4+ minutes)

- Bounce rate (my target: below 35%)
- Customer satisfaction score from website usability (my target: 4.5/5)

1.3 Current Website Analysis

I currently operate a basic brochure-style website that I built on a generic template back in 2019.

Strengths:

- I have a clean, memorable domain name (urbannest.co.za)
- My existing site has basic property listing functionality
- My contact form captures essential lead information

Weaknesses:

- My site is not mobile-responsive; I have noticed that 70% of my traffic bounces immediately on mobile devices
- I have no property search or filtering capabilities — users must browse page by page
- I cannot offer virtual tours; users must physically visit every property
- My application process is entirely manual, requiring PDF downloads and email submissions
- I have no tenant portal for my existing renters
- My load times are slow (average 6.8 seconds) due to unoptimized images and shared hosting

Areas for Improvement:

- I need a complete rebuild with a mobile-first responsive design
- I want to implement advanced search with map integration
- I need to add virtual 360° tour capabilities
- I must create automated online application workflows
- I plan to build a secure tenant portal for maintenance requests and rent payments

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1.4 Proposed Website Features and Functionality

Essential Features:

1. **Homepage**
    - A hero section with a prominent search bar (location, price range, bedrooms)
    - A featured properties carousel showcasing my best listings
    - Trust indicators (reviews, properties I manage, cities I cover)
    - Quick-links to popular neighborhoods where I operate
2. **Property Search & Listings**
    - An interactive map-based search with pin clustering
    - Advanced filters (price, bedrooms, pet-friendly, parking, furnished)
    - Sort options (price low-high, newest, most popular)
    - Grid and list view toggle
3. **Property Detail Pages**
    - A high-resolution image gallery with zoom and fullscreen
    - Embedded 360° virtual tours for my key properties
    - Interactive floor plans
    - Neighborhood insights (walkability score, nearby amenities)
    - "Schedule Viewing" and "Apply Now" CTAs
4. **Online Application System**
    - A step-by-step digital application form
    - Document upload (ID, proof of income, references)
    - Real-time application status tracking so applicants know where they stand
    - Digital signature integration
5. **Tenant Portal**
    - A secure login dashboard for my current tenants
    - A rent payment gateway
    - Maintenance request submission with photo upload
    - Lease document access and renewal reminders

6. **About Us & Contact**
    - My team profiles and company story
    - An FAQ section addressing common tenant questions
    - A live chat widget during my business hours
    - A contact form with Google Maps integration showing my office

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1.5 Design and User Experience

Colour Scheme:

- **Primary:** Deep Navy (#1A237E) — I chose this to convey trust, professionalism, and stability
- **Secondary:** Warm Coral (#FF6B6B) — I use this for CTAs and highlights to add energy and action
- **Accent:** Soft Gold (#FFD54F) — I selected this for a premium feel on featured listings and badges
- **Neutrals:** Off-White (#FAFAFA) backgrounds, Charcoal (#333333) body text, Light Grey (#E0E0E0) borders

Typography:

- **Headings:** Aptos Bold, sizes 32px (H1), 24px (H2), 18px (H3)
- **Body Text:** Aptos Regular, 16px with 1.6 line-height for readability
- **Captions/Labels:** Aptos Regular, 14px, uppercase with letter-spacing for tags

Layout and Design Approach:

- I want a mobile-first responsive grid system
- Card-based layout for property listings with generous white space
- A sticky navigation bar with search always accessible
- Progressive disclosure — advanced filters hidden behind "More Filters" to reduce cognitive load
- Skeleton screens during loading to improve perceived performance

User Experience Considerations:

- **Navigation:** I want a persistent top bar with a hamburger menu on mobile; breadcrumb trails on listing pages

- **Search:** Auto-suggest locations as users type; "Recent Searches" and "Saved Searches" for returning users
- **Accessibility:** I am committed to WCAG 2.1 AA compliance — keyboard navigation, ARIA labels, alt text for all images, minimum 4.5:1 contrast ratios
- **Performance:** Lazy loading for images, pagination for listings, compressed assets

Low-Fidelity Wireframes:

plain

```
[Logo]  Search...   [Login] [Menu]  ← Sticky Nav
                        |
  Find Your Perfect Urban Home
Location | Price | Beds |          ← Hero Search
                        |
[Filter: Price] [Beds] [Type ]
                        |
 | IMG  |       | IMG  |      | IMG  |
 | |R2,500/m| |R3,200/m| |R4,100/m|    | ← Property Cards
 | |2 Bed   | |1 Bed   | |Studio |    |
 | [Load More] Page 1 of 12            |

 | [Footer: About | Contact | Legal]  |
```

1.6 Technical Requirements

Hosting & Domain:

- I will retain my existing domain: urbannest.co.za
- I plan to use AWS EC2 (t3. medium instance) with CloudFront CDN for global asset delivery
- SSL Certificate: Let's Encrypt (auto-renewing)

- Database: MongoDB Atlas (M10 cluster) for flexible property schema and scalability

Programming Languages & Frameworks:

- **Frontend:** Next.js 14 (React framework) with TypeScript for type safety
- **Styling:** Tailwind CSS for utility-first responsive design
- **Backend:** Node.js with Express.js API
- **Database:** MongoDB (NoSQL for property documents), Redis for session caching
- **Search:** Algolia for instant search and filtering
- **Media:** Cloudinary for image optimization and 360° tour hosting
- **Payments:** Pay Fast integration for South African rent payments
- **Maps:** Google Maps Platform API (Places, Maps JavaScript, Geocoding)

1.7 Timeline and Milestones

Table

Phase	Duration	Milestone	Deliverable
Phase 1: Discovery & Planning	Week 1–2	Requirements finalized	Project brief, sitemap, user personas
Phase 2: Design	Week 3–4	Design approval	High-fidelity mockups, design system, wireframes
Phase 3: Development Sprint 1	Week 5–7	Core functionality	Homepage, search, listing pages, CMS integration
Phase 4: Development Sprint 2	Week 8–10	Advanced features	Tenant portal, application system, payment gateway

Phase	Duration	Milestone	Deliverable
Phase 5: Testing & QA	Week 11–12	Bug-free build	Cross-browser testing, mobile testing, security audit
Phase 6: Deployment	Week 13	Site live	Production deployment, DNS cutover, monitoring setup
Phase 7: Post-Launch	Week 14–16	Optimization	Performance tuning, analytics review, staff training

Total Project Duration: 16 weeks (4 months)

1.8 Budget

Table

Category	Estimated Cost (ZAR)	Notes
Design & UX	R18,000	UI/UX design, wireframes, prototyping
Frontend Development	R35,000	Next.js, responsive implementation
Backend Development	R28,000	API, database, integrations
Content Creation	R8,000	Professional photography for 20 of my properties

Category	Estimated Cost (ZAR)	Notes
Third-Party Services	R12,000	Algolia, Google Maps API, Cloudinary (annual)
Hosting & Infrastructure	R6,000/year	AWS EC2, MongoDB Atlas, CDN
Domain & SSL	R500/year	.co.za renewal + SSL
Testing & QA	R7,000	Cross-device testing, security audit
Project Management	R5,000	Client meetings, documentation
Contingency (10%)	R11,950	Buffer for scope changes
TOTAL	**R131,450**	One-time + first-year operational costs

1.9 References

Proposal 2: Haven Hills Vacation Rentals

2.1 Organisation Overview

Name: Haven Hills Vacation Rentals

History: Haven Hills began in 2015 when the Morrison family converted their ancestral farmhouse in the Drakensberg foothills into a holiday retreat. Word-of-mouth success led them to partner with neighbouring landowners, and today Haven Hills represents 28 unique countryside properties — from rustic log cabins to luxury stone cottages —

across the KwaZulu-Natal Midlands and Eastern Cape Highlands. The brand celebrates slow living, local heritage, and reconnection with nature.

Mission Statement: To curate unforgettable countryside escapes by connecting travelers with authentic, character-filled properties while supporting local communities and preserving natural landscapes.

Vision Statement: To be the leading platform for sustainable, experience-driven rural tourism in Southern Africa.

Target Audience: Primary: Families with children aged 5–16 seeking active holidays (35–50 years old). Secondary: Couples aged 28–45 seeking romantic weekend getaways. Tertiary: Remote workers seeking "workation" retreats. Average household income: R40,000–R100,000 annually. These users value authenticity, detailed information, and immersive visual storytelling before booking.

2.2 Website Goals and Objectives

Specific Goals:

- *Increase direct bookings by 50% (reducing reliance on third-party platforms like Airbnb and Booking.com)*
- *Build an email community of 5,000 subscribers within 12 months through content marketing*
- *Showcase each property's unique story and surrounding experiences*
- *Improve seasonal occupancy from 45% to 70% through targeted promotions and packages*

Key Performance Indicators (KPIs):

- *Direct booking conversion rate (target: 5% of site visitors)*
- *Average booking value (target: R4,500 per stay)*
- *Email newsletter sign-up rate (target: 3% of visitors)*
- *Return visitor rate (target: 25%)*
- *Organic search traffic growth (target: 20% month-over-month for first 6 months)*
- *Social media referral traffic (target: 15% of total traffic)*

2.3 Current Website Analysis

Haven Hills currently has no dedicated website. The business operates entirely through Airbnb, Booking.com, and a Facebook page with 1,200 followers. This creates significant vulnerabilities and misses opportunities.

Strengths (Current Channels):

- *Strong guest reviews (4.8/5 average across platforms)*
- *Beautiful property photography already exists*
- *Established brand recognition in the Midlands region*
- *Existing booking calendar data and availability management*

Weaknesses:

- *No owned digital asset — platform algorithm changes directly impact revenue*
- *15–20% commission fees per booking paid to third-party platforms*
- *No ability to capture guest emails for repeat marketing*
- *Limited storytelling capability; listings are transactional, not experiential*
- *No upselling of add-ons (guided hikes, farm-to-table dinners, spa treatments)*
- *No search engine visibility for "Drakensberg cottages" or related terms*

Areas for Improvement:

- *Build a branded, bookable website to own the customer relationship*
- *Implement a content strategy (blog, guides, seasonal itineraries)*
- *Create package offerings that combine accommodation + experiences*
- *Establish direct payment processing to eliminate commission fees*
- *Integrate with existing channel manager to prevent double bookings*

2.4 Proposed Website Features and Functionality

Essential Features:

1. *Immersive Homepage*
    - *Full-screen video hero showing aerial Drakensberg scenery*
    - *"Find Your Escape" mood-based search (e.g., "Family Adventure," "Romantic Retreat," "Digital Detox")*
    - *Seasonal highlights and weather widgets*

- *Trust signals: guest reviews, sustainable tourism badges, local partnership logos*

2. *Property Discovery*
    - *Grid layout with large hero images and subtle hover effects*
    - *Filter by region, group size, pet-friendly, accessibility, hot tub/fireplace/Wi-Fi*
    - *"Compare Properties" side-by-side tool*
    - *Interactive map showing properties relative to hiking trails, waterfalls, and towns*
3. *Property Detail Pages*
    - *Photo gallery with lightbox and caption storytelling*
    - *Property "character profile" (history, unique features, host story)*
    - *Amenities checklist with icons*
    - *Availability calendar with dynamic pricing*
    - *Guest reviews with verified stay badges*
    - *"Add Experiences" upsell section before checkout*
4. *Experience Packages*
    - *Curated add-ons: Guided fly-fishing, horseback riding, farm tours, private chefs*
    - *Seasonal packages: "Winter Fireside Escape," "Spring Wildflower Trail"*
    - *Gift voucher purchases*
5. *Booking Engine*
    - *Real-time availability with instant confirmation*
    - *Multi-currency display (ZAR, USD, GBP, EUR)*
    - *Deposit payment (50% upfront) or full payment options*
    - *Booking modification and cancellation portal*
    - *Automated pre-arrival emails with directions and local tips*
6. *Journal (Blog)*
    - *Seasonal travel guides, packing lists, and local stories*

- *SEO-optimized content targeting long-tail keywords*
- *Email capture embedded in articles*

7. *About & Sustainability*
    - *Host stories and property histories*
    - *Environmental commitments (solar power, water conservation, local sourcing)*
    - *Community impact report*

2.5 Design and User Experience

Colour Scheme:

- *Primary: Forest Green— nature, growth, sustainability*
- *Secondary: Burnt Sienna— earth, warmth, rustic charm*
- *Accent: Cream— parchment, comfort, heritage*
- *Neutrals: Stone Grey for secondary text, Warm White backgrounds, Deep Brown for headings*

Typography:

- *Headings: Aptos Bold with occasional serif accent for property names (Playfair Display) to evoke heritage*
- *Body Text: Aptos Regular, 16px, 1.7 line-height for relaxed reading*
- *Quotes/Stories: Aptos Italic, 18px, for host testimonials and property histories*

Layout and Design Approach:

- *Editorial, magazine-style layouts rather than rigid grids*
- *Generous whitespace with nature photography as the dominant visual element*
- *Asymmetric compositions for property cards to feel curated, not catalog-like*
- *Sticky booking bar on property pages that follows scroll*
- *Seasonal colour variations (lighter greens in spring, deeper oranges in autumn)*

User Experience Considerations:

- *Navigation: Simple top nav with "Properties," "Experiences," "Journal," "About," "Contact"*

- *Mood-Based Browsing: Instead of forcing date/location first, allow users to browse by "vibe" or experience type*
- *Trust: Prominent review scores, host verification badges, and secure payment icons*
- *Mobile: Thumb-friendly date pickers, click-to-call buttons, and offline-ready PDF guides*
- *Accessibility: Alt text describing scenery for screen readers, skip-to-content links, focus indicators*

Low-Fidelity Wireframes:

plain

[Logo] Properties Journal About 🛒]

|

│ [Full-bleed Nature Video] │ ← Hero

|

│ Escape to the Hills │

|

│ [Family] [Romance] [Solo] │

|

│ Handpicked Countryside Retreats │

│ [IMG] ││ [IMG] │

│ The Old Mill ││ Cedar Cabin │

│ From R1,200 ││ From R2,400 │

│ ★4.9 (28) ││ ★4.8 (41) │

[Journal: Winter Guide] ← Content

Read More →

|

[Newsletter: Get seasonal offers]

[Footer: Sustainability | FAQ | Terms]

2.6 Technical Requirements

Hosting & Domain:

- *Domain: havenhills.co.za (to be registered)*
- *Hosting: SiteGround or local South African provider (Afrihost) for data sovereignty and speed*
- *CDN: Cloudflare for image delivery and DDoS protection*
- *SSL: Let's Encrypt wildcard certificate*

Programming Languages & Frameworks:

- *CMS: WordPress 6.5+ with custom theme (PHP, HTML5, CSS3, JavaScript)*
- *Booking Engine: WooCommerce Bookings plugin with custom extensions*
- *Page Builder: Gutenberg blocks with custom ACF (Advanced Custom Fields) for property profiles*
- *Search: ElasticPress for fast property filtering*
- *Maps: Mapbox GL JS (more affordable than Google Maps for high traffic)*

Phase	*Duration*	*Milestone*	*Deliverable*
Phase 1: Discovery	*Week 1–2*	*Strategy alignment*	*Content audit, competitor analysis, sitemap*
Phase 2: Content & Photography	*Week 3–4*	*Content ready*	*Property descriptions, host interviews, image optimization*
Phase 3: Design	*Week 5–6*	*Design sign-off*	*Homepage, property page, booking flow mockups*
Phase 4: Development	*Week 7–10*	*Feature-complete build*	*WordPress theme, booking engine, integrations*

Phase	*Duration*	*Milestone*	*Deliverable*
Phase 5: Content Population	*Week 11*	*All properties live*	*28 properties with full descriptions and calendars*
Phase 6: Testing	*Week 12*	*Launch-ready*	*Payment testing, mobile testing, booking flow QA*
Phase 7: Launch & Training	*Week 13*	*Sites live*	*Staff training on CMS, booking management, analytics*
Phase 8: Marketing Setup	*Week 14–16*	*Growth-ready*	*GA4, Mailchimp automation, basic SEO optimization*

- *Payments: PayFast and Stripe for credit card and Instant EFT*
- *Email: Mailchimp API integration for newsletter and automated guest communications*
- *Channel Manager API: Integration with existing Beds24 or like sync calendars with Airbnb/Booking.com*
- *Analytics: Google Analytics 4 + Microsoft Clarity for heatmaps*

2.7 Timeline and Milestones

Table

Total Project Duration: 16 weeks (4 months)

2.8 Budget

Table

Category	*Estimated Cost (ZAR)*	*Notes*
Domain Registration	*R200/year*	*.co.za domain*
Hosting (Premium)	*R4,800/year*	*Managed WordPress with SSL*
WordPress Theme Development	*R25,000*	*Custom theme, responsive, ACF*
WooCommerce & Booking Setup	*R15,000*	*Custom booking rules, deposits*
Third-Party Integrations	*R8,000*	*Mapbox, Mailchimp, channel manager API*
Content Migration & Entry	*R6,000*	*28 properties, blog posts, pages*
Professional Copywriting	*R5,000*	*Property stories, SEO content*
Photography (Supplemental)	*R4,000*	*Drone footage and host portraits*
Payment Gateway Setup	*R2,500*	*PayFast + Stripe configuration*
Testing & QA	*R4,000*	*Cross-browser, booking flow testing*
Training & Documentation	*R3,000*	*Staff manuals, video tutorials*
Contingency (10%)	*R7,750*	*Scope adjustments*

Category	*Estimated Cost (ZAR)*	*Notes*
TOTAL	*R85,250*	*One-time + first-year operational*

2.9 References